Supplementary material for

"Leakage-controlled evaluation of stacking and recursive feature elimination for diabetes outcome prediction on the PIMA dataset"

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S1 Validation design: where each loop runs

Figure S1 shows the complete validation design. There are three nested levels. The *outer* level is the 5×10-fold repeated cross-validation used for model comparison. Within each outer training part there are two *inner* levels: the 5-fold cross-validation that RFECV uses to choose the subset size, and the 5-fold split that produces out-of-fold probabilities for the stacking meta-learner. The 154-patient hold-out set is separated before any modelling and is used once, by the final pipeline refitted on all 614 development patients.

S2 Statistical testing procedure

Repeated cross-validation Let x_{rk} be the value of a metric on fold k of repeat r ($r = 1, \ldots, 5$; $k = 1, \ldots, 10$; $J = 50$). The point estimate is $\bar{x} = J^{-1} \sum_{r,k} x_{rk}$. Fold estimates share most of their training data, so the naive variance s^2/J is too small. Following Nadeau and Bengio, we use

$$\widehat{\mathrm{Var}}(\bar{x}) = \left(\frac{1}{J} + \frac{n_{\mathrm{test}}}{n_{\mathrm{train}}} \right) s^2, \qquad \frac{n_{\mathrm{test}}}{n_{\mathrm{train}}} = \frac{1}{9}, \tag{S1}$$

with s^2 the sample variance of the J fold estimates. 95% confidence intervals are $\bar{x} \pm t_{0.975,\, J-1} \sqrt{\widehat{\mathrm{Var}}}$.

Paired comparisons For a configuration A and the reference B (logistic regression on all features), $d_{rk} = x^A_{rk} - x^B_{rk}$ is computed on identical folds. The corrected resampled statistic is $t = \bar{d} / \sqrt{(1/J + 1/9)\, s_d^2}$, referred to a t distribution with $J - 1 = 49$ degrees of freedom (two-sided). The analysis comprises 23 configurations: 11 learners × 2 feature sets (all 16 predictors or the common RFECV subset) = 22, plus logistic regression with its own LR-ranked RFECV. Each of the 22 non-reference configurations is compared with the reference (logistic regression on all 16 predictors), and the 22 p-values are adjusted with the Holm step-down procedure. This comparison set was fixed in the analysis code before the repeated cross-validation was run.

Why this test The corrected resampled t-test is designed for exactly this setting, where fold estimates from repeated k-fold cross-validation are correlated. In simulation studies it keeps the type I error close to the nominal level and replicates well across random partitions, whereas the uncorrected paired t-test is anti-conservative. Alternatives such as a nested bootstrap of the whole pipeline or Bayesian correlated t-tests are valid. However, a nested bootstrap would have required thousands of refits of RFECV and the stack (about 45 s each), and the conclusions here are far from any decision boundary: the stacking–logistic-regression difference is +0.001 with an adjusted $p = 1.00$.

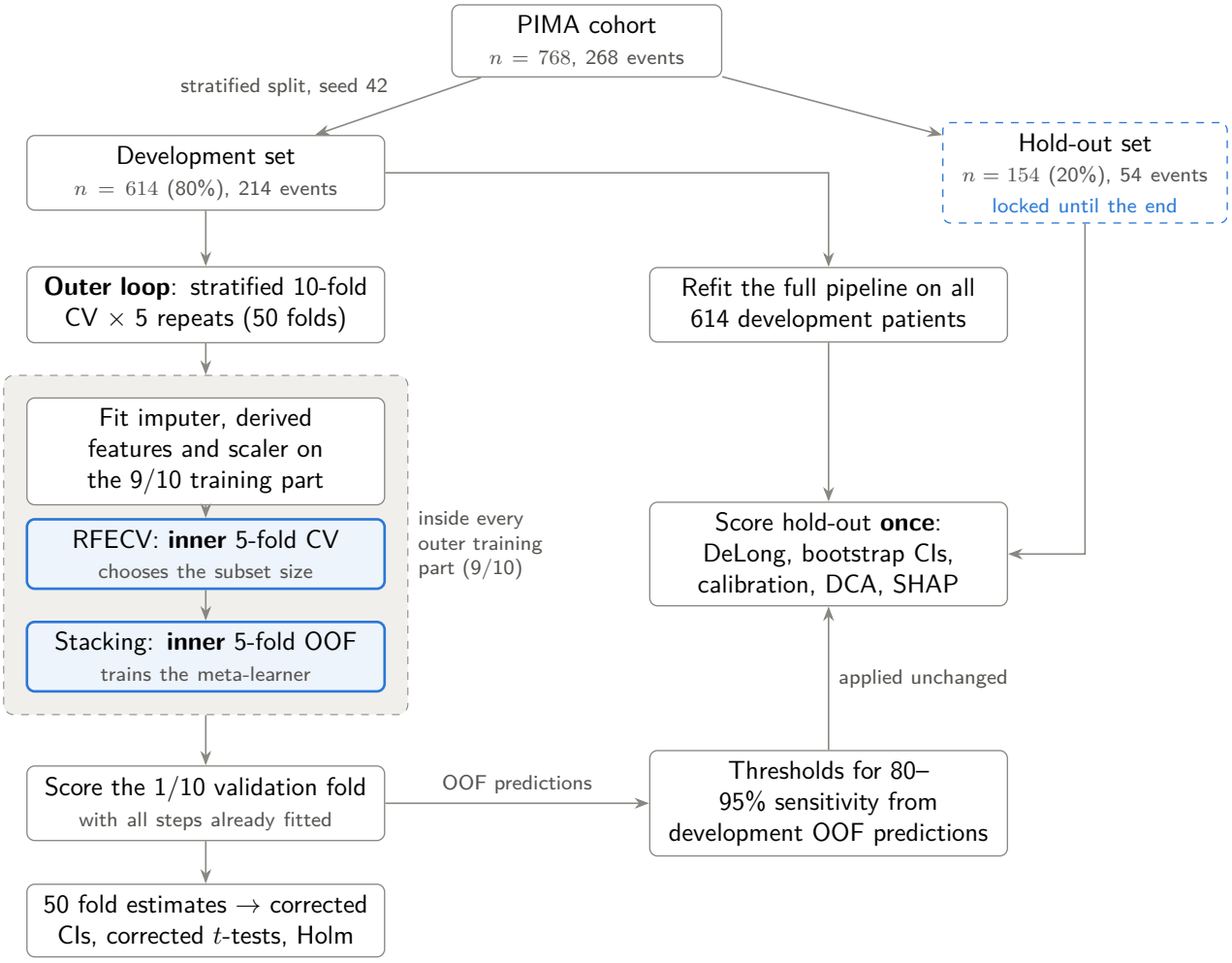


Figure S1: Nested validation design. Blue boxes mark the two inner resampling loops that run inside every outer training part. Nothing from an outer validation fold or from the hold-out set is used to fit any step.

Hold-out set ROC-AUC confidence intervals and pairwise comparisons on the 154 hold-out patients use DeLong's non-parametric covariance (fast mid-rank implementation), with Holm adjustment over six comparisons against stacking + RFECV. All other metrics use 95% percentile bootstrap intervals from 2000 resamples of hold-out patients; resamples containing a single class are discarded.

Calibration and thresholds The calibration intercept is estimated from $\operatorname{logit} P(y = 1) = a + \operatorname{logit} \hat{p}$ (offset model) and the slope from $\operatorname{logit} P(y = 1) = a' + b \operatorname{logit} \hat{p}$, both by Newton–Raphson maximum likelihood. For a target sensitivity s, the threshold is the largest value t for which at least $\lceil s\, n_+ \rceil$ of the $n_+ = 214$ development-set positives have an out-of-fold probability (averaged over the five repeats) of at least t, i.e. the $\lceil s\, n_+ \rceil$-th largest positive probability. Patients with probability $\geq t$ are flagged, so ties at the threshold are always flagged. The pooled out-of-fold calibration analysis uses all 3070 predictions (614 unique patients $\times$ 5 repeats). The development prevalence is $214/614 = 0.349$ (logit $= -0.626$).

S3 Full cross-validated results

Table S1: All 23 configurations from 5×10-fold repeated cross-validation on the development set ($n = 614$). Means with Nadeau–Bengio corrected 95% CIs beneath. "+ RFECV" is the common random-forest-ranked RFECV subset. The last two columns give the unadjusted and Holm-adjusted corrected resampled t-test p-values for the ROC-AUC difference from logistic regression on all features. Threshold-based metrics use 0.5.

Model	Feat.	Acc.	Bal. acc.	Prec.	Sens.	Spec.	F1	MCC	ROC-AUC	Brier	p	p_{Holm}
Stacking	16.0	0.759 (0.719–0.799)	0.760 (0.722–0.797)	0.635 (0.579–0.691)	0.763 (0.710–0.815)	0.757 (0.700–0.814)	0.689 (0.646–0.732)	0.505 (0.431–0.580)	0.846 (0.812–0.880)	0.164 (0.144–0.183)	0.586	1.000
Stacking + RFECV	12.0	0.756 (0.714–0.797)	0.755 (0.715–0.795)	0.631 (0.574–0.688)	0.753 (0.697–0.810)	0.757 (0.700–0.814)	0.684 (0.638–0.730)	0.497 (0.418–0.575)	0.843 (0.809–0.878)	0.165 (0.144–0.185)	0.890	1.000
Logistic regression	16.0	0.764 (0.727–0.801)	0.759 (0.725–0.794)	0.646 (0.592–0.700)	0.744 (0.690–0.798)	0.775 (0.721–0.828)	0.688 (0.647–0.729)	0.507 (0.438–0.576)	0.842 (0.809–0.875)	0.163 (0.144–0.183)	–	–
Logistic regression + RFECV	12.0	0.760 (0.721–0.798)	0.750 (0.713–0.786)	0.647 (0.588–0.706)	0.717 (0.664–0.770)	0.782 (0.727–0.838)	0.676 (0.633–0.720)	0.492 (0.418–0.566)	0.842 (0.811–0.873)	0.163 (0.145–0.182)	0.997	1.000
LR (LR-ranked RFECV)	10.1	0.760 (0.725–0.796)	0.754 (0.717–0.790)	0.641 (0.591–0.692)	0.731 (0.671–0.791)	0.776 (0.728–0.824)	0.680 (0.637–0.724)	0.496 (0.426–0.567)	0.840 (0.808–0.873)	0.164 (0.145–0.183)	0.487	1.000
SVM	16.0	0.775 (0.735–0.815)	0.747 (0.705–0.790)	0.695 (0.624–0.767)	0.655 (0.587–0.723)	0.840 (0.790–0.889)	0.670 (0.614–0.726)	0.505 (0.418–0.591)	0.840 (0.802–0.878)	0.156 (0.138–0.175)	0.796	1.000
SVM + RFECV	12.0	0.765 (0.722–0.808)	0.736 (0.693–0.780)	0.680 (0.605–0.755)	0.643 (0.578–0.708)	0.830 (0.776–0.884)	0.656 (0.599–0.713)	0.483 (0.393–0.572)	0.837 (0.800–0.873)	0.158 (0.140–0.176)	0.577	1.000
Extra trees + RFECV	12.0	0.753 (0.713–0.794)	0.739 (0.698–0.780)	0.642 (0.581–0.703)	0.693 (0.627–0.759)	0.786 (0.732–0.840)	0.662 (0.612–0.712)	0.474 (0.393–0.556)	0.835 (0.798–0.872)	0.161 (0.142–0.179)	0.429	1.000
Extra trees	16.0	0.753 (0.707–0.798)	0.735 (0.690–0.780)	0.647 (0.576–0.717)	0.676 (0.609–0.743)	0.794 (0.736–0.852)	0.656 (0.601–0.712)	0.469 (0.377–0.560)	0.835 (0.795–0.874)	0.160 (0.140–0.181)	0.448	1.000
Naive Bayes + RFECV	12.0	0.762 (0.725–0.799)	0.730 (0.692–0.768)	0.682 (0.607–0.756)	0.624 (0.564–0.683)	0.836 (0.787–0.886)	0.646 (0.595–0.698)	0.473 (0.393–0.554)	0.832 (0.797–0.867)	0.205 (0.174–0.235)	0.263	1.000
Random forest	16.0	0.756 (0.710–0.802)	0.746 (0.698–0.793)	0.641 (0.575–0.707)	0.712 (0.639–0.785)	0.780 (0.724–0.836)	0.671 (0.613–0.728)	0.484 (0.390–0.578)	0.831 (0.795–0.868)	0.164 (0.144–0.184)	0.269	1.000
Random forest + RFECV	12.0	0.759 (0.715–0.804)	0.750 (0.705–0.794)	0.645 (0.580–0.711)	0.718 (0.652–0.785)	0.781 (0.725–0.838)	0.676 (0.622–0.730)	0.492 (0.403–0.581)	0.831 (0.797–0.866)	0.164 (0.145–0.184)	0.262	1.000
Gradient boosting	16.0	0.746 (0.709–0.784)	0.711 (0.670–0.752)	0.657 (0.589–0.726)	0.595 (0.520–0.669)	0.828 (0.779–0.876)	0.618 (0.562–0.675)	0.436 (0.353–0.518)	0.831 (0.798–0.863)	0.164 (0.145–0.183)	0.294	1.000
Naive Bayes	16.0	0.773 (0.739–0.807)	0.739 (0.703–0.775)	0.701 (0.636–0.766)	0.626 (0.566–0.686)	0.852 (0.808–0.895)	0.657 (0.608–0.706)	0.494 (0.420–0.568)	0.830 (0.793–0.867)	0.205 (0.175–0.235)	0.159	1.000
Gradient boosting + RFECV	12.0	0.748 (0.709–0.788)	0.714 (0.671–0.757)	0.660 (0.588–0.733)	0.601 (0.525–0.677)	0.827 (0.776–0.878)	0.623 (0.564–0.682)	0.441 (0.354–0.528)	0.830 (0.798–0.862)	0.164 (0.145–0.183)	0.291	1.000
KNN	16.0	0.760 (0.718–0.802)	0.726 (0.681–0.770)	0.681 (0.605–0.757)	0.611 (0.541–0.680)	0.841 (0.790–0.891)	0.639 (0.578–0.700)	0.466 (0.375–0.558)	0.824 (0.781–0.866)	0.162 (0.141–0.183)	0.108	1.000
XGBoost	16.0	0.744 (0.701–0.787)	0.713 (0.670–0.757)	0.650 (0.576–0.723)	0.611 (0.543–0.678)	0.816 (0.761–0.871)	0.625 (0.567–0.683)	0.436 (0.346–0.527)	0.821 (0.784–0.857)	0.173 (0.149–0.196)	0.056	0.954
KNN + RFECV	12.0	0.757 (0.713–0.800)	0.721 (0.674–0.769)	0.675 (0.600–0.750)	0.604 (0.525–0.683)	0.839 (0.788–0.889)	0.632 (0.567–0.697)	0.458 (0.361–0.554)	0.821 (0.777–0.865)	0.163 (0.143–0.183)	0.110	1.000
XGBoost + RFECV	12.0	0.746 (0.707–0.785)	0.716 (0.674–0.757)	0.651 (0.583–0.719)	0.615 (0.543–0.687)	0.817 (0.764–0.869)	0.627 (0.572–0.682)	0.440 (0.357–0.524)	0.820 (0.785–0.854)	0.172 (0.150–0.193)	0.033	0.586
LightGBM	16.0	0.731 (0.687–0.775)	0.706 (0.660–0.752)	0.620 (0.551–0.690)	0.622 (0.546–0.698)	0.789 (0.733–0.846)	0.617 (0.556–0.677)	0.414 (0.321–0.508)	0.803 (0.764–0.842)	0.205 (0.172–0.237)	0.002	0.044
LightGBM + RFECV	12.0	0.733 (0.691–0.775)	0.709 (0.667–0.750)	0.625 (0.559–0.691)	0.627 (0.559–0.695)	0.790 (0.733–0.848)	0.621 (0.568–0.674)	0.420 (0.336–0.505)	0.800 (0.765–0.836)	0.202 (0.175–0.230)	<0.001	0.020
Decision tree + RFECV	12.0	0.710 (0.668–0.752)	0.723 (0.681–0.765)	0.565 (0.517–0.613)	0.767 (0.702–0.832)	0.679 (0.626–0.733)	0.648 (0.601–0.696)	0.429 (0.348–0.510)	0.772 (0.731–0.813)	0.207 (0.180–0.235)	<0.001	0.020
Decision tree	16.0	0.710 (0.669–0.751)	0.725 (0.683–0.766)	0.564 (0.518–0.610)	0.774 (0.708–0.839)	0.675 (0.622–0.729)	0.650 (0.604–0.696)	0.431 (0.352–0.511)	0.770 (0.729–0.811)	0.210 (0.181–0.239)	0.001	0.025

S4 Full hold-out results

Table S2: All hold-out metrics ($n = 154$, 54 events) at a 0.5 threshold. 95% CIs beneath: DeLong for ROC-AUC, percentile bootstrap (2000 resamples) otherwise. AP, average precision; Int., calibration intercept; Slope, calibration slope.

Model	Acc.	Bal. acc.	Prec.	Sens.	Spec.	F1	MCC	ROC-AUC	Brier	AP	Int.	Slope
Stacking + RFECV	0.766	0.777	0.629	0.815	0.740	0.710	0.532	0.822	0.177	0.696	-0.624	0.866
	(0.701–0.831)	(0.711–0.842)	(0.516–0.736)	(0.709–0.915)	(0.649–0.823)	(0.615–0.797)	(0.403–0.658)	(0.756–0.888)	(0.143–0.213)			
Stacking	0.747	0.758	0.606	0.796	0.720	0.688	0.494	0.819	0.177	0.702	-0.623	0.865
	(0.675–0.812)	(0.690–0.826)	(0.493–0.712)	(0.688–0.904)	(0.631–0.804)	(0.592–0.778)	(0.362–0.625)	(0.752–0.885)	(0.143–0.213)			
Logistic regression	0.740	0.740	0.606	0.741	0.740	0.667	0.464	0.811	0.181	0.683	-0.629	0.756
	(0.675–0.805)	(0.670–0.811)	(0.491–0.722)	(0.625–0.862)	(0.649–0.825)	(0.564–0.763)	(0.325–0.606)	(0.744–0.878)	(0.147–0.219)			
LR + LR-ranked RFECV	0.734	0.731	0.600	0.722	0.740	0.655	0.447	0.809	0.182	0.684	-0.634	0.746
	(0.662–0.805)	(0.660–0.806)	(0.480–0.714)	(0.608–0.846)	(0.649–0.825)	(0.551–0.753)	(0.309–0.591)	(0.741–0.876)	(0.147–0.220)			
Random forest + RFECV	0.734	0.723	0.607	0.685	0.760	0.643	0.434	0.826	0.166	0.711	-0.338	0.776
	(0.662–0.799)	(0.648–0.794)	(0.482–0.724)	(0.564–0.804)	(0.673–0.840)	(0.537–0.739)	(0.290–0.573)	(0.761–0.892)	(0.134–0.201)			
XGBoost + RFECV	0.760	0.734	0.660	0.648	0.820	0.654	0.470	0.824	0.171	0.697	+0.039	0.559
	(0.688–0.825)	(0.660–0.805)	(0.528–0.781)	(0.523–0.776)	(0.743–0.888)	(0.543–0.752)	(0.321–0.605)	(0.758–0.891)	(0.132–0.215)			
SVM	0.727	0.684	0.630	0.537	0.830	0.580	0.383	0.810	0.169	0.684	+0.004	0.844
	(0.656–0.799)	(0.609–0.759)	(0.489–0.765)	(0.407–0.672)	(0.755–0.896)	(0.457–0.692)	(0.226–0.534)	(0.740–0.880)	(0.135–0.205)			

S5 Feature-selection frequencies

Table S3: Percentage of the 50 outer folds in which each candidate predictor was retained by the common random-forest-ranked RFECV and by coefficient-ranked RFECV for logistic regression. Derived features are deterministic functions of the original variables.

Predictor	Type	RF-ranked (%)	LR-ranked (%)
`Pregnancies`	original	58	94
`Glucose`	original	100	100
`BloodPressure`	original	68	2
`SkinThickness`	original	56	0
`Insulin`	original	46	26
`BMI`	original	100	82
`DiabetesPedigreeFunction`	original	94	92
`Age`	original	68	94
`Glucose_BMI`	derived	100	78
`Glucose_Age`	derived	100	84
`HOMA_IR_surrogate`	derived	100	72
`Insulin_Glucose_Ratio`	derived	68	80
`BMI_Age`	derived	100	38
`Obese`	derived	6	74
`IGT_2h`	derived	38	26
`Pedigree_Age`	derived	100	64

S6 TRIPOD+AI reporting map

Table S4 maps the main TRIPOD+AI reporting domains (Collins et al., BMJ 2024;385:e078378) to the manuscript. The domain descriptions are paraphrased. Before submission, the author should complete the official 27-item checklist, available from the TRIPOD website, using this map.

Table S4: TRIPOD+AI domains and where they are addressed (main text section or this supplement).

Domain (paraphrased)	Where addressed
Title and structured abstract	Title; Abstract
Background, rationale and objectives	Sections 1–2
Data source, setting and participants	Section 3.1
Outcome definition and timing	Section 3.1 (Outcome definition and intended setting); Section 5.6
Predictors, including derived features	Section 3.1; Section 3.3; Table 2
Sample size	Section 3.1; Section 5.6 (limitation)
Missing data	Section 3.2; Section 3.9; Section 4.6; Section 5.4; Table 8
Data preparation and leakage control	Section 3; Fig. 1; Fig. S1
Model types, hyperparameters and building	Sections 3.4–3.5; Table 3
Class imbalance handling	Section 3.2 (balanced class weights); Section 4.4 (calibration effect)
Internal validation design	Section 3.6; Fig. S1; Section S2
Performance measures (discrimination, calibration, utility)	Sections 3.7–3.8; Section S2
Fairness across subgroups	Not assessed: single-sex, single-ethnicity cohort (Section 5.6)
Results: participants and model specification	Section 3.1; Section 4; Tables S1–S3
Results: performance with uncertainty	Tables 4–8; Tables S1–S2
Interpretation, limitations and usability	Section 5
Explainability	Section 3.10; Section 4.7; Fig. 7
Ethics, funding and conflicts	Declarations
Protocol, registration, patient and public involvement	Not applicable or not done (retrospective analysis of public data)
Data and code availability	Data and code availability statement; Section S8

S7 Configurations compared

Table S5: The 23 configurations of the repeated cross-validation. "Common RFECV" is the random-forest-ranked subset fitted once per outer fold and shared by all learners. Settings of each learner are given in Table 3 of the main text.

Learner	In stack	Feature sets	Role
Logistic regression	yes	all 16; common RFECV; LR-ranked RFECV	reference (all 16); learner-specific RFECV check
SVM (RBF, Platt)	yes	all 16; common RFECV	base learner
k-NN	yes	all 16; common RFECV	base learner
Random forest	yes	all 16; common RFECV	base learner
Extra trees	yes	all 16; common RFECV	base learner
Gradient boosting	yes	all 16; common RFECV	base learner
XGBoost	yes	all 16; common RFECV	base learner
LightGBM	yes	all 16; common RFECV	base learner
Gaussian naive Bayes	no	all 16; common RFECV	reference learner
Decision tree (depth 5)	no	all 16; common RFECV	reference learner
Stacking (LR meta-learner)	–	all 16; common RFECV	ensemble under study
Total: 11 learners × 2 feature sets = 22, plus LR with LR-ranked RFECV = 23 configurations.			

S8 Software environment and reproduction

All results were produced by commit `e23102d` of `https://github.com/subrahmanyamtanala-rgb/T2D` with Python 3.11, scikit-learn 1.9.1, XGBoost 3.2.0, LightGBM 4.7.0, SHAP, NumPy, pandas and SciPy, using random seed 42 throughout. The dataset is included in the repository as `data/diabetes.csv`. It was obtained from the public mirror `https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv`, a header-less copy of the 768-record UCI PIMA Indians Diabetes Database, and a header row with the standard column names was added. The file has 768 records and 268 positive outcomes, and its SHA-256 checksum is `3c7023e0935097ac63e03c6f398928b6dfce30974d5591d3af09a9661876d9fe`. Before submission, the author should confirm that it matches the UCI or Kaggle distribution, which have the same record and class counts. To reproduce:

```
pip install -r requirements.txt shap
python -m t2d.analysis --stage cv        # ~40 min on 4 CPU cores
python -m t2d.analysis --stage holdout   # ~7 min
python -m t2d.analysis --stage missing   # ~35 min
paper/build.sh                           # all tables, figures and PDFs
```

Fold-level metrics, out-of-fold predictions, hold-out predictions and SHAP values are provided in `results/revision/`.

Reproducibility check Commit `881d84f` only renamed the derived feature `Hyperglycemic` to `IGT_2h`; the computation depends on column order, not names. To check reproducibility and the rename, the hold-out stage was re-run in a fresh output directory at that commit. All hold-out predictions, metrics, bootstrap intervals, DeLong tests, thresholds, calibration estimates, decision curves and SHAP summaries matched the reported values to within floating-point rounding (maximum absolute difference 2.2×10^{-16}). The 50-fold cross-validation and missing-data stages were not re-run for this check.

Random seeds Every stochastic operation uses seed 42:

- the stratified 80/20 split (`train_test_split`);
- the repeated outer folds (`RepeatedStratifiedKFold`, 10 folds $\times$ 5 repeats; 10×2 for the missing-data analysis);
- the inner 5-fold splits of RFECV and of the stacking meta-learner (`StratifiedKFold`, shuffled);
- the RFECV random-forest ranker and every learner with a random component (LR, SVM, decision tree, random forest, extra trees, gradient boosting, XGBoost, LightGBM);
- the iterative imputer;
- the bootstrap resampling (NumPy `default_rng`);
- the 40-patient SHAP background sample (`DataFrame.sample`).

The Platt calibration of the SVM uses deterministic, unshuffled internal folds. k-NN, naive Bayes and median or k-NN imputation are deterministic.

S9 NHANES 2013–2014 replication cohort

Source The official CDC data files could not be downloaded from the analysis environment. The cohort was built from the public repository `https://github.com/ChintanaL/diabetes-prediction-using-nhanes-dataset` (commit `074e664`), which contains processed extracts of the 2013–2014 public-use files: `labs_new.csv` (OGTT glucose `LBXGLT`, insulin `LBXIN`, HbA1c `LBXGH`), `demographic_new.csv` (sex, age) and `examination_full.csv` (BMI, diastolic blood pressure). Respondent identifiers range from 73557 to 83731, the 2013–2014 cycle.

Removal of imputed values The repository's author had imputed missing laboratory and examination values. Such values do not match NHANES' published resolution: for example, 280 glucose values equal 115.5503… and 39 insulin values equal 12.4658…. We kept glucose and diastolic blood pressure only if integer, insulin only with at most two decimals, and HbA1c and BMI only with at most one decimal; all other values were set to missing. Of 1454 laboratory records, 1144 had genuine glucose, insulin and HbA1c values, and 902 of these were aged ≥ 21 years. BMI was missing for 3 and blood pressure for 30 participants; both were imputed inside the validation folds.

Cohort and outcome 902 adults (456 women), 288 with HbA1c $\geq 5.7\%$ (31.9%). The derived file `data/nhanes_2013_2014_ogtt.csv` has SHA-256 `957039e34efbd8a8c0965f14790cd2b2dbb83df8609433b08df0313`

Regenerating from the official files `python -m t2d.nhanes -cdc <dir>` builds the same cohort from the CDC files `DEMO_H`, `BMX_H`, `BPX_H`, `OGTT_H`, `INS_H` and `GHB_H` (diastolic pressure as the mean of the non-zero readings). The analysis is then re-run with `python -m t2d.analysis -dataset nhanes -stage cv` and `-stage holdout`. Results are in `results/external_nhanes/`.